Journal of Ethnopharmacology 131 (2010) 485–496

Contents lists available at ScienceDirect

Journal of Ethnopharmacology

journal homepage: www.elsevier.com/locate/jethpharm

The analgesic and anti-inflammatory effect of WIN-34B, a new herbal formula for osteoarthritis composed of *Lonicera japonica* Thunb and *Anemarrhena asphodeloides* BUNGE *in vivo*

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ARTICLE INFO

Article history:
Received 25 January 2010
Received in revised form 29 June 2010
Accepted 9 July 2010
Available online 17 July 2010

Keywords:
Lonicera japonica Thunb
Anemarrhena asphodeloides BUNGE
Anti-nociceptive activity
Anti-inflammatory activity
WIN-34B
Osteoarthritis

ABSTRACT

Ethnopharmacological relevance: : *Lonicera japonica* Thunb and *Anemarrhena asphodeloides* BUNGE have been used for the treatment of a variety of inflammatory diseases, cold and infective diseases in many countries, including Korea and China.

Aim of the study: : This study aimed to assess the anti-nociceptive and anti-inflammatory activities of *n*-butanol fraction (WIN-34B) prepared from dried flowers of *Lonicera japonica* and dried roots of *Anemarrhena asphodeloides* as potential novel treatment of osteoarthritis.

Materials and methods: : Anti-nociceptive activities of WIN-34B (100, 200 and 400 mg/kg, p.o.) were measured using acetic acid-induced writhing response, formalin-induced paw licking, hot plate, radiant heat tail-flick, carrageenan-induced paw pressure, and Hargreaves tests, respectively. Anti-inflammatory activities of WIN-34B (100, 200 and 400 mg/kg, p.o.) were assessed using acetic acid-induced vascular permeability, carrageenan-induced paw edema, and croton oil-induced ear edema. Anti-osteoarthritis effect of WIN-34B was analyzed using monosodium iodoacetate (MIA)-induced osteoarthritis animal model.

Results: : WIN-34B exhibited better anti-inflammatory activity than that of celecoxib in carrageenan at the dose of 200 mg/kg and croton oil-induced paw edema and ear edema at the doses of 200 and 400 mg/kg. WIN-34B exhibited significant anti-inflammatory effects on vascular permeability. WIN-34B also exhibited significant anti-nociceptive activities in the late phase of formalin-induced paw licking and writhing response model in mice. In radiant heat tail-flick and carrageenan-induced paw pressure tests, WIN-34B at the dose of 400 mg/kg and at the doses of 200 and 400 mg/kg presented similar activities to indomethacin and celecoxib. Compared to indomethacin WIN-34B at 400 mg/kg showed similar or better anti-nociceptive activities after 1 and 2 h of theraphy in the hot plate test and better anti-nociceptive activity than that of celecoxib in Hargreaves test. In the MIA-induced osteoarthritis animal models, WIN-34B at 400 mg/kg exhibited similar or better anti-nociceptive property than that of celecoxib throughout the pain measurement periods.

Conclusion: : When compared to celecoxib, WIN-34B exhibited similar or better anti-nociceptive and anti-inflammatory activities in osteoarthritic animal models, which may become a potential novel treatment for osteoarthritis.

1. Introduction

Osteoarthritis (OA) is the most common joint disease and an important cause of physical disability in the elderly (Ameye and Chee, 2006). Symptoms include joint pain, stiffness, limited movement, joint deformity, and varying degrees of joint inflammation (Lark et al., 1997; Poole, 1999). Current therapeutic strategies for OA focus on alleviating symptoms, particularly due to pain and inflammation. Non-steroidal anti-inflammatory drugs (NSAIDs) and corticosteroids present the mainstream of treatments for OA (Kidd, 2006). Although NSAIDs are recommended as an initial drug therapy to reduce joint inflammation and pain, their

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doi:10.1016/j.jep.2010.07.025

Table 1
The known components of flowers of *Lonicera japonica* Thunb.

Alpha-pinene(+,−)	Loganin
Alpha-terpineol	Lonicerin
Apigenin	Loniceroside A
Auroxanthin	Loniceroside B
Benzyl alcohol	Luteolin
Beta-carotene (all-trans)	Lycopene
Caffeic acid methyl ester	2-phenylethanol
Carvacrol	Phytofluene
Chlorogenic acid	Secologanin
Choline	Secologanin dimethylacetal
cis-2,6,6-trimethyl-2-vinyl-5-hydroxytetrahydropyran	Secoxyloganin
Epivogeloside	Tauroside G3
Epsilon-carotene	Trans-2-methyl-2-vinyl-5-(alpha-hydroxyisopropyl)-tetrahydrofuran
Gamma-carotene	Trans-2-methyl-2-vinyl-5-(alpha-hydroxy-isopropyl)-tetrahydrofuran
Geraniol	Vanillic acid
Hederagenin-3-O-[alpha-L-rhamnopyranosyl-(1->2)-alpha-L-arabinopyranosyl]-28-O-beta-D-xylopyranosyl-(1->6)-beta-D-glucopyranosyl ester	Venoterpine
Hederagenin-3-O-[alpha-L-rhamnopyranosyl(1->2)-alpha-L-arabinopyranosyl]-28-O-[3-O-acetyl-beta-D-glucopyranosyl-(1->6)-beta-D-glucopyranoside	Vogeloside
O-[3-O-acetyl-beta-D-xylopyranosyl-(1->6)-beta-D-glucopyranoside	Zeaxanthin (all E-form)
Linalool	

This table was referred from www.tradimed.com.

chronic use is limited by gastrointestinal-related toxicities, including nausea, dyspepsia, upper gastrointestinal bleeding, and ulcer perforation (Wolfe and Cathey, 1991). To minimize such toxicities in patients, better tolerated NSAIDs, such as the COX-2 inhibitors, have been used recently. However, their cardiovascular risks, including myocardial infarction and stroke, have been reported to be drug induced (Hippisley-Cox and Coupland, 2005; Lenzer, 2005; Pratico and Dogne, 2005; Tegeder and Geisslinger, 2006). This has led to the removal of rofecoxib from the market and to the issuance of recommendations limiting the use of celecoxib (Grosser et al., 2006). Corticosteroids have immunosuppressive and anti-inflammatory properties. Although they provide effective symptomatic relief, their substantial toxicities limit long-term use (Saag, 1997). In addition, the therapeutic approach is not curative, but rather to relieve clinical signs and symptoms of the disease. For curative treatment of osteoarthritis, efforts have been made to develop a more effective and safe drug. Joins^{TM}, a combination herbal drug containing the extracts of *Clematis mandshurica*, *Trichosanthes kirilowii* and *Prunella vulgaris*, commonly used for the treatment of osteoarthritis in Korea, is an example of such curative treatment developments for osteoarthritis. Clinical studies have demonstrated that Joins^{TM} relieves joint pain and improves functionality in osteoarthritis patients. The mechanism behind its efficacy may be attributed to cartilage protection and anti-inflammation (Hartog et al., 2008). However, its deficiency in immediate analgesic effect is a major drawback. In order to develop a better therapeutic alternative to Joins^{TM}, the following investigations were conducted with 200 medicinal herbs used in Chinese medicine, which has been widely used for the treatment of inflammatory diseases such as lymphadenitis and arthritis in Far Asia: cartilage protection in human osteoarthritic explant culture, anti-inflammatory effects in RAW 267.3 cells *in vitro*, anti-nociceptive in writhing, formalin, paw pressure tests, anti-inflammatory effects in paw edema and ear edema *in vivo*. Two herbs, namely, flowers of *Lonicera japonica* Thunb and roots of *Anemarrhena asphodeloides* BUNGE demonstrated excellent anti-nociceptive and anti-inflammatory properties in several animal models, as well as cartilage protection and anti-inflammatory properties in *in vitro* assays from our preliminary data. Thereafter, these two herbs were selected and mixed into a 2:1 (w/w) formulation, *i.e.* WIN-34B. The flower buds of *Lonicera japonica* Thunb (LJ, Caprifoliaceae) are commonly used in traditional Chinese medicine for the treatment of various diseases, including arthritis, diabetes mellitus, fever, infections, sores, and swelling. They are also consumed in indigenous beverages in Korea and China. Pharmacological studies have shown that extracts of *L. japonica* (LJ) flower buds have a broad spectrum of biological activity, including antibacterial, anti-inflammatory, anti-nociceptive, antioxidant, antiangiogenic, antipyretic, antiviral, and hepatoprotective effects (Qian et al., 2007, 2008). It is known that LJ contains various compounds (see Table 1). Various compounds such as chlorogenic acid, isochlorogenic acid, rutin and lonicerin, had been isolated and identified. Compounds such as chlorogenic acid, caffeic acid, rutin, lonicerin and 4,5-di-*O*-caffeoyl quinic acid were regarded as the major active biological components in LJ (Qian et al., 2008). LJ has also secoiridoid glucodises such as loganin and secologanin (Tomassini et al., 1995). Luteolin has been known as an active main constituent of LJ and suppresses inflammatory mediator released by blocking NF-κB and MAPKs activation pathway in HMC-11 cells (Kang et al., 2010). Roots of *Anemarrhena asphodeloides* BUNGE (AA, family Liliaceae) have been used as an antipyretic, anti-inflammatory, antidiabetic, anti-platelet aggregator and antidepressant in traditional medicine in China, Japan, and Korea. It is known that AA contains various compounds (see Table 2). AA primarily contains mangiferin and steroidal saponins such as sarsasapogenin and timosaponin AII and

Table 2
The known components of roots of *Anemarrhena asphodeloides* BUNGE.

Anemaran A	cis-Monomethylhinokiresinol
Anemaran B	cis-Oxyhinokiresinol
Anemaran C	cis-Tetrahydrohinokiresinol
Anemaran D	F-Gitonin
Anemarrhenasaponin I	Isomangiferin
Anemarrhenasaponin II	Mangiferin
Anemarrhenasaponin III	Nicotinamide
Anemarrhenasaponin IV	Nicotinic acid
Anemarsaponin A2	Parigenin
Anemarsaponin A1	p-Hydroxyphenyl crotonic acid
Anemarsaponin B	Timosaponin A-III
Anemarsaponin F	Timosaponin B-II
Anemarsaponin G	trans-Hinokiresinol
cis-Diacetylhinokiresinol	2,6,4′-Trihydroxy-4-methoxybenzophenone
cis-Hinokiresinol	

This table was referred from www.tradimed.com.

BII (Jung et al., 2009). The present study was conducted in rodents to investigate the analgesic and anti-inflammatory activities of WIN-34B for treatment of osteoarthritis by comparing its activities with that of commonly used anti-osteoarthritic drugs, including Joins^{TM}, celecoxib or indomethacin.

2. Materials and methods

2.1. Plant material

Dried flowers of *Lonicera japonica* and dried root of *Anemarrhena asphodeloides* were purchased from Song Lim Pharmaceutical Company (Seoul, Korea). They were identified by the Korea Pharmaceutical Trading Association (Seoul, Korea). Voucher specimen of *Lonicera japonica* Thunb (No. OA-LOJ-15) and *Anemarrhena asphodeloides* BUNGE (No. OA-ANA-11) were deposited at Central Research Institute, WhanIn Pharm., Co., Ltd (Suwon, Korea).

2.2. Preparation of n-butanol fraction prepared from dried flowers of Lonicera japonica and dried roots of Anemarrhena asphodeloides

Dried flowers of *Lonicera japonica* and dried roots of *Anemarrhena asphodeloides* having common characterization of UPLC patterns purchased from various sources based on UPLC fingerprintings of them are chosen. 2:1 (w/w) mixture of the two medicinal herbs, 100 g of such chosen dried flowers of *Lonicera japonica* and 50 g of such chosen dried roots of *Anemarrhena asphodeloides* were extracted with 1.35 l of 50% (v/v) ethanol in water for 4 h at 85 °C. After the extracted solution was filtered and evaporated *in vacuo*, the resulting concentrate was dissolved in 225 ml of distilled water and partitioned with 195 ml of *n*-butanol. The *n*-butanol layer was evaporated *in vacuo* and lyophilized for complete removal of the residual solvent to a yield brown powder of 11 g, with a yield of 7%. The final resultant extract was called WIN-34B.

2.3. HPLC analysis of chlorogenic acid and mangiferin contents of WIN-34B

WIN-34B was standardized based on the content of chlorogenic acid, a known major constituent of *Lonicera japonica* Thunb and mangiferin, and a known major constituent of *Anemarrhena asphodeloides* BUNGE. Chromatographic analysis of WIN-34B was performed with a reverse-phase High Performance Liquid Chromatography (HPLC) Waters equipped with the Waters Breeze System (2695 Separations Module, 996 Photodiode Array Detector, Empower 2 Software Build 2154). Separation was carried out using YMC Hydrosphere C_{18} column (4.6 × 250 mm, particle diameter of 5 μm, YMC, Kyoto, Japan) at 30 °C. The mobile phase was consisted of 0.1% phosphoric acid solution in pump A (88.8%) and acetonitrile in pump B. Elution was undertaken using step gradients (B; 11.2–11.2% (0–16 min), 11.2–13.2% (16–17 min), 13.2–13.2% (17–28 min), and 13.2–100% (28–33 min)) at a flow rate of 1.0 ml/min. Detections of chlorogenic acid and mangiferin were performed at 327 and 254 nm, respectively.

2.4. UPLC analysis of constituents of WIN-34B

UPLC analysis of WIN-34B was performed with a reverse-phase Ultra Performance Liquid Chromatography (UPLC) Waters equipped with the Waters Breeze System (AcQuity^{TM} Ultra Performance LC, Photodiode Array Detector, Empower 2 Software Build 2154). Separation was carried out using ACQITY UPLC® BEH C_{18} column (2.1 × 100 mm, particle diameter of 1.7 μm, ACQITY, Ireland) at 30 °C. The mobile phase was consisted of 0.1% Trifluoroacetic acid in water solution in pump A (95.0%) and 0.1% Trifluoroacetic acid in acetonitrile sloution in pump B. Elution was undertaken using step gradients (B; 5.0–10.0% (0–16 min), 10.0–25.0% (16–35 min), 25.0–50.0% (35–40 min), and 50.0–100% (40–47 min)) at a flow rate of 0.23 ml/min. Detections for constituents of WIN-34B were performed at 254 nm. Constituents of WIN-34B were identified by known standards and LC-MS/MS analysis.

2.5. Experimental animals

Male ICR mice, Wistar/ST rats and Spague Dawley (SD) rats were obtained from OrientBio Co., Ltd. (Sungnam, Korea), Japan SLC (Shizuoka, Japan) and Central Lab. Animal Inc. (Seoul, Korea). Male ICR mice (12–15, 18–20, 18–21, 19–25 and 21–24 g) were used in croton oil-induced ear edema, hot plate, acetic acid-induced vascular permeability, acetic acid-induced writing response, and tail-flick tests. Wistar/ST rats (130–150 g) were used in carrageenan-induced paw pressure and paw edema. SD rats (140–170, 190–220 g) were used in MIA-induced osteoarthritis and formalin-induced paw licking tests.

They were housed in standard cages at a constant temperature of 22 ± 1 °C, a relative humidity 55 ± 5% with a 12 h alternate light–dark cycle (08:00–20:00) for at least 1 week prior to the experiment. Animals used in this study were housed and cared for in accordance with the *NIH Guidelines for the Care and Use of Laboratory Animals*. The experimental protocol was approved by the Committee of Ethics on Animal Research, Central Research Institute, WhanIn Pharm. Co. Ltd. All tests were conducted under the guidelines of the *International Association for the Study of Pain* (Zimmermann, 1983).

2.6. Acute oral toxicity

To evaluate the acute toxicities of WIN-34B after a single oral dose, 20 male and 20 female SD rats were assigned randomly to two experimental groups: the vehicle (0.5% CMC in distilled water) group and the WIN-34B (1.25, 2.5, 5 g/kg) groups. Each experimental group was administered orally with the vehicle (0.5% CMC in distilled water) and WIN-34B (1.25, 2.5, 5 g/kg). Each group consisted of 5 male and 5 female. Mortality, clinical signs, body weight changes and gross findings were monitored for 14 days after treatment. This study was conducted in ChemOn Co., Ltd., a GLP institute approved by the Korea Food and Drug Administration (KFDA), in compliance with the *Testing Guidelines for the Safety Evaluation of Drugs* (Notification No.2005-60) issued by the KFDA and OECD GLP Principles of Good Laboratory Practice (1997).

2.7. Acetic acid-induced writhing response

The anti-nociceptive activities of WIN-34B were assessed by measuring the response to an intraperitoneal injection of acetic acid solution, which presented as contraction of the abdominal muscles and stretching of the hind limbs (Fischer et al., 2008). Each experimental group was administered orally with the vehicle (0.5% CMC in distilled water), WIN-34B (100, 200 or 400 mg/kg), Joins^{TM} (400 mg/kg) or celecoxib (100 mg/kg). One hour after oral administration of one of the investigational agents, 1% acetic acid at the dose of 10 ml/kg body weight was injected. After 5 min, the number of writhes was recorded during the subsequent 20 min period.

2.8. Formalin test

This test was based on the method used in the previous study (Shimoyama et al., 1997). Fifty microliter of 5% formalin was injected subcutaneously into the right hind paw of the rat. The time (in seconds) that the animal spent licking in response to the injected paw served as an indicator of pain response. Responses were measured for 10 min after formalin injection (early phase) and between 10 and 60 min after formalin injection (late phase). Each experimental group was administered orally with the vehicle (0.5% CMC in distilled water), WIN-34B (100, 200 or 400 mg/kg), Joins (400 mg/kg) or indomethacin (30 mg/kg) 30 min before formalin injection.

2.9. Hot plate test

The hot plate test was conducted at a fixed temperature of $55 \pm 1\,^{\circ}C$ according to the method previously reported (Hu et al., 2008). The reaction consisted of hind leg flinching, paw licking and jumping. The time in seconds between reaching the platform and the reaction provoked was recorded as the response latency. Mice exhibiting latency time between 3 and 8 s were chosen. The latency time was determined at 60 and 120 min after administration of the test samples, vehicle (0.5% CMC in distilled water), WIN-34B (100, 200 or 400 mg/kg), Joins^{TM} (400 mg/kg) or indomethacin (30 mg/kg). If the reaction time was more than 60 s, the latency was recorded as 60 s.

2.10. Tail-flick test

The tail-flick test was carried out in accordance with the method described elsewhere (Zvejniece et al., 2006), with minor modification. In brief, the tail of the mouse was placed on the photo element window of a Tail-flick analgesia meter (Model No. 336, IITC, 23924 Victory Blvd Woodland Hills, CA 91367, USA) and an infrared beam was focused on the tail area, 3 cm from its base. The latency time to respond to pain stimuli was recorded. To avoid tissue damage, maximal exposure to pain stimuli was restricted to 15 s. The tail-flick response was tested 60 min after administration of the test samples, as follows: vehicle (0.5% CMC in distilled water), WIN-34B (100, 200 or 400 mg/kg), Joins^{TM} (400 mg/kg) or indomethacin (30 mg/kg).

2.11. Carrageenan-induced paw pressure

The model of inflammatory nociception was adapted from previous reported method (Randall and Selitto, 1957). Rats (150–180 g) were injected with carrageenan at $t = -4$ h. The effects of WIN-34B on carrageenan-induced inflammation were assessed using an analgesia-meter (Ugo Basile, Comerio, Italy). This device generated a mechanical force on the affected paw and the nociceptive threshold was defined as the force (ing) at which the rat withdraws the paw. The behavioral endpoint (paw withdrawal) and the effect of WIN-34B thereupon were assessed. After adapting to the analgesia-meter for 3 days, animals were randomly allocated to different treatment groups according to their body weights; vehicle (0.5% CMC in distilled water), WIN-34B (100, 200 or 400 mg/kg), Joins^{TM} (400 mg/kg) or celecoxib (100 mg/kg). 1 mg carrageenan in 0.1 ml saline (carrageenan 1%) was injected subcutaneously into the left hind paw of the rat to induce inflammation 1 h after administering test samples to the rats in each group. Inflammation-induced pain was measured by the analgesia-meter. Inhibition of carrageenan-induced hypernociception for the treatments was presented as the nociceptive threshhold, expressed in grams (g). The nociceptive threshold (g) was the force applied to the dorsal surface of the inflamed hind paw, which caused each animal to withdraw its paw.

2.12. Hargreaves test

Thermal hyperalgesia was assessed in unrestrained rats using a procedure adapted from previously published reports (Hargreaves et al., 1988). Rats (n = 10–11 per group) were placed in opaque, plastic chambers ($13 \times 24 \times 13$ cm) positioned on a glass surface. Animals were allowed to habituate in this environment for 20 min prior to testing. Paw withdrawal latency in response to radiant heat was measured using the plantar test apparatus (Ugo Basile, Comerio, Italy). The heat source was positioned beneath the plantar surface of the affected hind paw before being activated. The digital timer connected to the heat source automatically recorded the response latency for paw withdrawal to the nearest tenth of a second. A cut-off time of 22 s was used to prevent tissue damage. The paw withdrawal latency of each rat was measured three times at each test interval and the median score recorded. Each experimental group was administered orally with the vehicle (0.5% CMC in distilled water), WIN-34B (100, 200 or 400 mg/kg), Joins^{TM} (400 mg/kg) or celecoxib (100 mg/kg). The rats received a 100 μl intraplantar injection of CFA (1 mg/ml heat-killed and dried *Mycobacterium tuberculosis*, each milliliter of vehicle containing 0.85 ml paraffin oil + 0.15 ml mannide monooleate) on the surface of the left hind paw 1 h after oral administration. The effects of WIN-34B on paw withdrawal latency were measured 3 and 24 h after CFA injection.

2.13. Acetic acid-induced vascular permeability

The acetic acid-induced vascular permeability test was carried out using a modification of the previously described method (Whittle, 1964). Thirty minutes after oral administration of vehicle (0.5% CMC in distilled water), WIN-34B (100, 200 or 400 mg/kg), Joins^{TM} (400 mg/kg) or celecoxib (100 mg/kg), 10 ml/kg body weight of 2% Evans blue was injected intravenously into each mouse tail. Twenty minutes later, 10 ml/kg body weight of 1% acetic acid in saline was injected intraperitoneally. Twenty minutes after the injection of acetic acid, the mice were sacrificed by cervical dislocation. After 5 ml of saline was injected into the abdominal cavity, the washings were collected into test tubes, then centrifuged at 2000 rpm for 10 min. The absorbance of the supernatant was read at 650 nm (A650) using an ELISA reader (Spectra max 190, Molecular Devices, Sunnyvale, USA) and the amounts of Evans blue leakage in abdominal cavity were measured from the absorbance measurement for the supernatant. Vascular permeability was represented as the amounts of Evans blue leakage in the abdominal cavities.

2.14. Carrageenan-induced paw edema

The anti-inflammatory activity of WIN-34B was determined by the carrageenan-induced edema test in rats' hind paws. One hundred microliters of 1% suspension of carrageenan in saline was injected into the plantar side of the left hind paws of the rats. WIN-34B at doses of 100, 200 and 400 mg/kg, Joins at the dose of 400 mg/kg, celecoxib at the dose of 100 mg/kg and the vehicle (0.5% CMC in distilled water) were administered orally (p.o.) 1 h before the carrageenan treatment. Paw volume was measured before carrageenan injection and at 3 h after the administration of the edematogenic agent using a plethysmometer (Ugo Basile, Comerio, Italy). The degree of swelling induced was evaluated by the ratio a/b, where a and b are volumes of the left hind paws after and before carrageenan treatment, respectively. Increase of paw volume (%) was calculated by the following equation; $(a - b)/b100$.

2.15. Croton oil-induced ear edema

Ear edema was induced by topical administration of croton oil (0.625 mg per ear, 25 μl) to the outer and inner surfaces of the right ear of each mouse. Each experimental group was administered orally with vehicle (0.5% CMC in distilled water), WIN-34B (100, 200 or 400 mg/kg), Joins^{TM} (400 mg/kg) or celecoxib (100 mg/kg) 1 h before administering the inflammatory agents. The left ear received the same volume of acetone (25 μl, vehicle). The change in the ear's

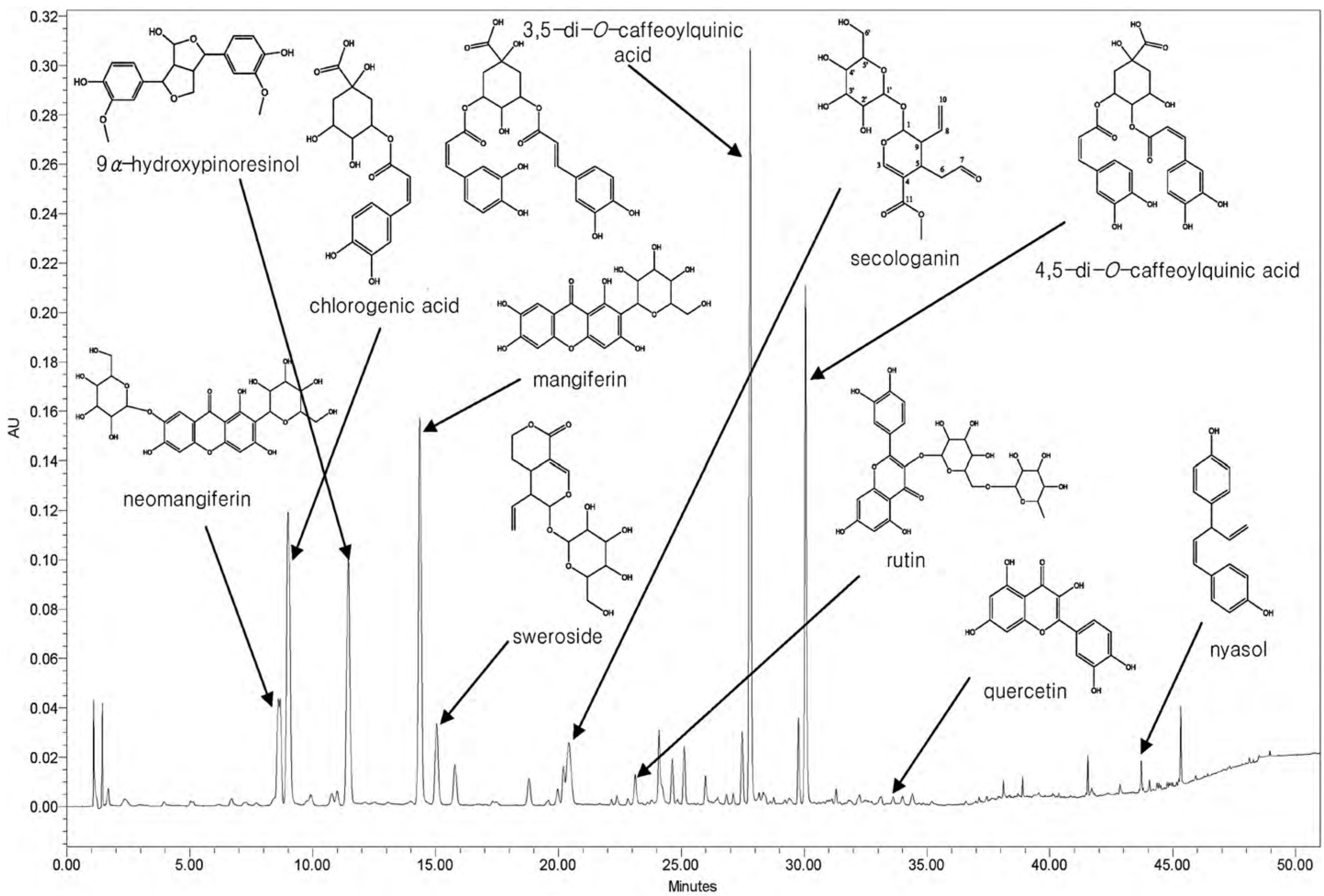

Fig. 1. UPLC fingerprints of WIN-34B at wavelength of 254.

thickness was measured with a thickness gauge (Model H, Peacock, Japan) 4 h after the administration of croton oil. The results were expressed as the difference in thickness between the inflamed right ear and the acetone-treated left ear. Increase in ear thickness (%) was attained as follows; $\{(\text{Right ear thickness} - \text{Left ear thickness})/\text{Left ear thickness}\} \times 100$.

2.16. MIA-induced osteoarthritis

Rats were anaesthetized with a Zoletil™ (anesthetics) and Rumpun™ (muscle relaxant) mixture, and were given a single intra-articular injection of monosodium iodoacetate (MIA) through the infrapatella ligament of the right knee. MIA was dissolved in 0.9% sterile saline and administered in a volume of 50 μl (60 mg/ml) using a 26 gauge, 0.5-inch needle. The effect of joint damage on weight distribution through the right (arthritic) and left (untreated) knee was assessed using an incapacitance tester (Linton Instrumentation, Norfolk, UK). Briefly, an incapacitance tester measures weight distribution on the two hind paws. The force exerted by each hind limb is measured in grams. Each experimental group was administered orally with vehicle (0.5% CMC in distilled water), WIN-34B (100, 200 or 400 mg/kg), Joins (400 mg/kg) or celecoxib (100 mg/kg) for 14 days. The effect of joint damage on weight distribution through the right (arthritic) and left (untreated) knee was measured at 4, 7, 11, 14 days after drug treatment.

2.17. Reagents and drugs

Chlorgenic acid (purity: 97%, reference standard) was purchased from USP (USP Rockville, MD, USA). Mangiferin (purity: 97%, reference standard) was purchased from ChromaDex (Irvine, USA). Methanol and *n*-butanol were purchased from Burdick & Jackson (Honeywell, USA). Acetic acid, Carboxymethyl Cellulose (CMC), monosodium iodoacetate (MIA), formalin, aceton, croton oil, carrageenan, celecoxib, and indomethacin were obtained from Sigma–Aldrich (St. Louis, USA). Zoletil™ was purchased from Virbac (Carros, France) and Rumpun™ was purchased from Bayer Korea (Seoul, Korea). CFA (Adjuvant complete freund) was purchased from Difco (Detroit, MI, USA). 0.9% saline was purchased from DAIHAN Pharm. Co. Ltd. (Seoul, Korea). Joins™ was purchased from SK Chemicals (Seoul, Korea). Deionized water from a Milli-Q system (Millipore, Bedford, MA, USA) was used to prepare all buffers and sample solutions.

2.18. Statistical analysis

Data are expressed as mean ± S.E.M. Statistical evaluation was carried out using one-way analysis of variance (ANOVA followed by Dunnett's post hoc test). Statistical differences were considered to be significant at $P < 0.05$. The program used for the statistical analysis was SigmaStat (version 11.0).

3. Results

3.1. The HPLC analysis of standard materials of WIN-34B

From the results of the standard calibration curve, the R^2 values of all marker substances ranged between 0.999 and 0.1000. The standard material used for the quantitative analysis of *Lonicera japonica* is chlorogenic acid (M.W. 345.31) and its content in

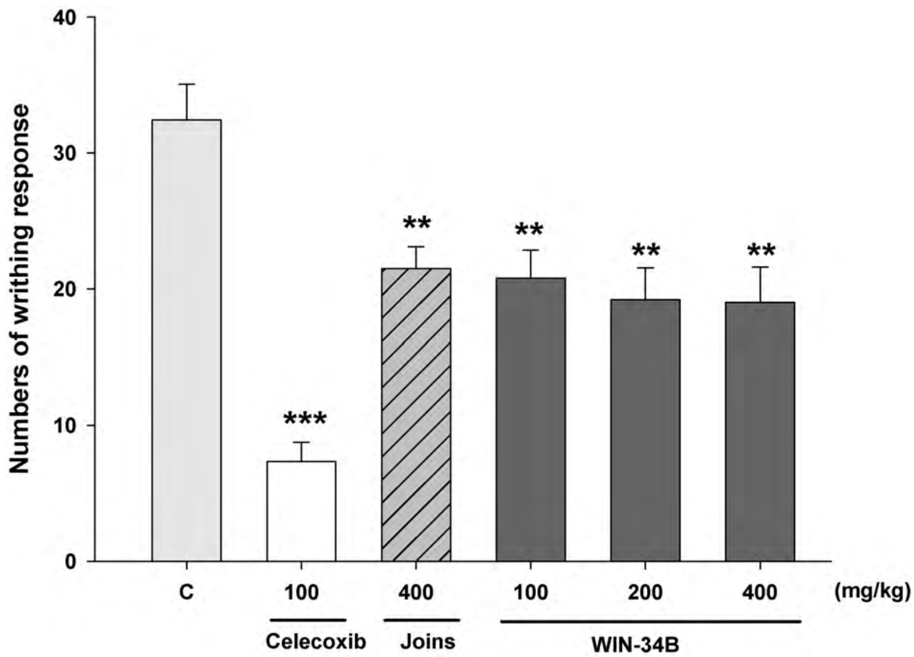

Fig. 2. Effects of WIN-34B on acetic acid-induced writhing response in mice. The number of writhes was measured after 1% acetic acid injection for 20 min. The values are expressed as the mean ± S.E.M. ($n = 5–8$). $^{*}p < 0.05$, $^{**}p < 0.01$ and $^{***}p < 0.001$ indicate statistically significant differences from the vehicle control group.

WIN-34B is 31.11 ± 0.12 mg/g (3.11 ± 0.38%). The standard material used for the quantitative analysis of *Anemarrhena asphodeloides* is mangiferin (M.W. 422.34) and its content in WIN-34B is 6.39 ± 0.03 mg/g (0.64 ± 0.52%). The WIN-34Bs having above percentages of 2.58 in chlorogenic acid and 0.52 in mangiferin are chosen. Such chosen WIN-34B presents above 90% efficacy of standard WIN-34B in writhing test and ear edema test. Finally, the chosen WIN-34B by such process was used in the present study.

3.2. UPLC analysis of constituents of WIN-34B

Several compounds of WIN-34B were identified (see Fig. 1). As seen in Fig. 1, neomangifrin, chlorogenic acid, 9α-hydroxypinoresinol, mangiferin, sweroside, secologanin, rutin, 3,5-di-O-caffeoylquinic acid, 4,5-di-O-caffeoylquinic acid, quecertin and nyasol in order of retention time were identified in WIN-34B by analyses of UPLC and Mass spectrometry. Chlorogenic acid, 9α-hydroxypinoresinol, sweroside, secologanin, rutin, 3,5-di-O-caffeoylquinic acid, 4,5-di-O-caffeoylquinic acid and quecertin may be originated from dried flowers of *Lonicera japonica*, while neomangiferin, mangiferin and nyasol may be originated from dried root of *Anemarrhena asphodeloides*.

3.3. Acute oral toxicity

To evaluate the acute oral toxicity of WIN-34B, we determined its three dose toxicities in both sexes of rats at 1.25, 2.5 and 5 g/kg body weight. These doses had no effect on mortality, body weight changes, gross findings, and clinical signs in both sexes (data not shown).

3.4. Acetic acid-induced writhing response

The cumulative amount of abdominal stretching correlated with the level of acetic acid-induced pain (Fig. 2). WIN-34B treatment (100, 200, and 400 mg/kg) significantly inhibited the number of writhes in comparison with normal controls in a dose-dependent manner. The inhibition rates of the number of writhes compared with the control are 35.9%, 40.8%, and 41.4%, respectively. This inhibiting effect of acetic acid-induced writhes by WIN-34B (400 mg/kg) was better than that produced by Joins^{TM} (400 mg/kg) with an inhibition rate of 33.7%.

3.5. Formalin test

WIN-34B significantly inhibited formalin-induced pain in the late phase (Table 3), however, there was no inhibition in the early phase (Table 3). The positive control indomethacin (30 mg/kg) also significantly ($p < 0.001$) inhibited formalin-induced pain in the late phase. Such inhibition of WIN-34B at a dose of 400 mg/kg showed better potency than that of Joins at dose of 400 mg/kg. WIN-34B treatment (200 and 400 mg/kg) significantly inhibited the formalin-induced pain (late phase) in comparison with normal controls. The inhibition rates of formalin-induced licking compared with control are 44.4% and 64.8%, respectively.

Table 3
Effects of WIN-34B on inflammatory pain induced by formalin in rats.

Group	Dose mg/kg, p.o.	Licking time (sec)	
		0–10 min	10–60 min
Control (0.5% CMC)	0	87.1 ± 17.7	156.6 ± 18.9
Indomethacin	30	58.7 ± 8.3	46.4 ± 11.1***
Joins	400	53.0 ± 10.1	84.0 ± 19.9*
	100	54.8 ± 7.7	118.5 ± 26.3
WIN-34B	200	73.7 ± 6.5	104.6 ± 8.5*
	400	61.7 ± 4.6	71.7 ± 6.0***

The values are expressed as the mean ± S.E.M. ($n = 6–8$).
* $p < 0.05$ indicates statistically significant differences from the vehicle control group.
*** $p < 0.001$ indicates statistically significant differences from the vehicle control group.

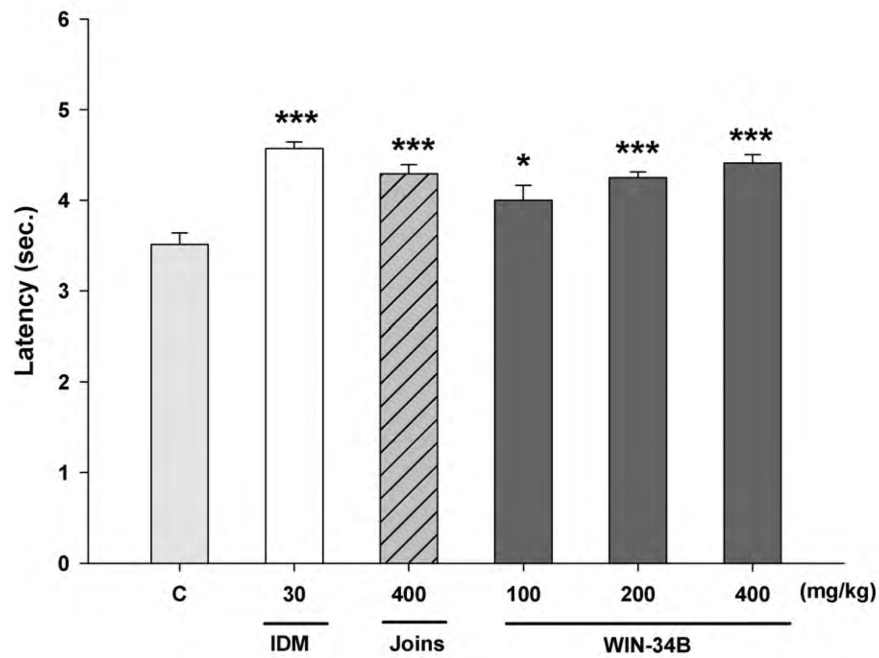

Fig. 3. Effects of WIN-34B on radiant heat tail-flick response in mice. The tail-flick response latency was measured when infrared beam was lighted on tail. The values are expressed as the mean ± S.E.M. ($n = 6–7$). $^{*}p < 0.05$, $^{**}p < 0.01$ and $^{***}p < 0.001$ indicate statistically significant differences from the vehicle control group.

3.6. Hot plate test

Hot plate test was also assayed to characterize the analgesic activity of WIN-34B. The results presented in Table 4 show that oral administration of WIN-34B at 200–400 mg/kg significantly increased the pain threshold at the observation time of 2 h in comparison with control ($p < 0.001$). WIN-34B at 400 mg/kg has higher increase of pain threshold compared to indomethacin. Also, WIN-34B at 400 mg/kg increased pain threshold significantly at the observation time of 1 h, while Joins^{TM} and indomethacin did not show any significant increase of pain threshold in comparison with control (see Table 4, $p < 0.05$). These results suggest that WIN-34B relieves pain faster and more effective than that of indomethacin.

3.7. Tail-flick test

The results in Fig. 3 represent that oral administration of WIN-34B at 100–400 mg/kg significantly increased the pain threshold in a dose-dependent manner, compared to control ($p < 0.001$). Also, WIN-34B at 400 mg/kg increased the latency as much as indomethacin.

3.8. Carrageenan-induced paw pressure

In Fig. 4, WIN-34B increased paw withdrawal threshold significantly ($p < 0.001$) in a dose-dependently manner. The effect of

Table 4
Effects of WIN-34B on hot plate reaction time in mice.

Group	Dose mg/kg, p.o.	Latency time (sec)	
		60 min	120 min
Control (0.5% CMC)	0	5.6 ± 0.5	4.5 ± 0.3
Indomethacin	30	6.6 ± 0.5	7.7 ± 0.6***
Joins	400	6.7 ± 0.6	8.1 ± 0.9**
	100	6.0 ± 0.7	6.8 ± 1.3
WIN-34B	200	6.8 ± 1.0	6.4 ± 0.7*
	400	7.8 ± 0.7*	7.8 ± 0.6***

The values are expressed as the mean ± S.E.M. ($n = 6–8$).
* $p < 0.05$ indicates statistically significant differences from the vehicle control group.
** $p < 0.01$ indicates statistically significant differences from the vehicle control group.
*** $p < 0.001$ indicates statistically significant differences from the vehicle control group.

Table 5
Effects of WIN-34B on CFA-induced thermal hyperalgesia assessed by Hargreves test in rats.

Group	Dose mg/kg, p.o.	Withdrawal latency time (sec)		
		0 h	3 h	24 h
Control (0.5% CMC)	0	11.5 ± 0.5	6.7 ± 0.7	4.8 ± 0.5
Celecoxib	100	11.9 ± 0.5	9.4 ± 0.8*	8.4 ± 0.8**
Joins	400	11.5 ± 0.5	8.2 ± 0.8	5.6 ± 0.6
	100	11.3 ± 0.5	7.8 ± 0.8	6.2 ± 0.5
WIN-34B	200	11.2 ± 0.4	9.7 ± 0.9*	7.9 ± 1.0*
	400	11.3 ± 0.3	10.4 ± 0.9**	9.1 ± 0.5***

The values are expressed as the mean ± S.E.M. ($n = 10–11$).
* $p < 0.05$ indicates statistically significant differences from the vehicle control group.
** $p < 0.01$ indicates statistically significant differences from the vehicle control group.
*** $p < 0.001$ indicates statistically significant differences from the vehicle control group.

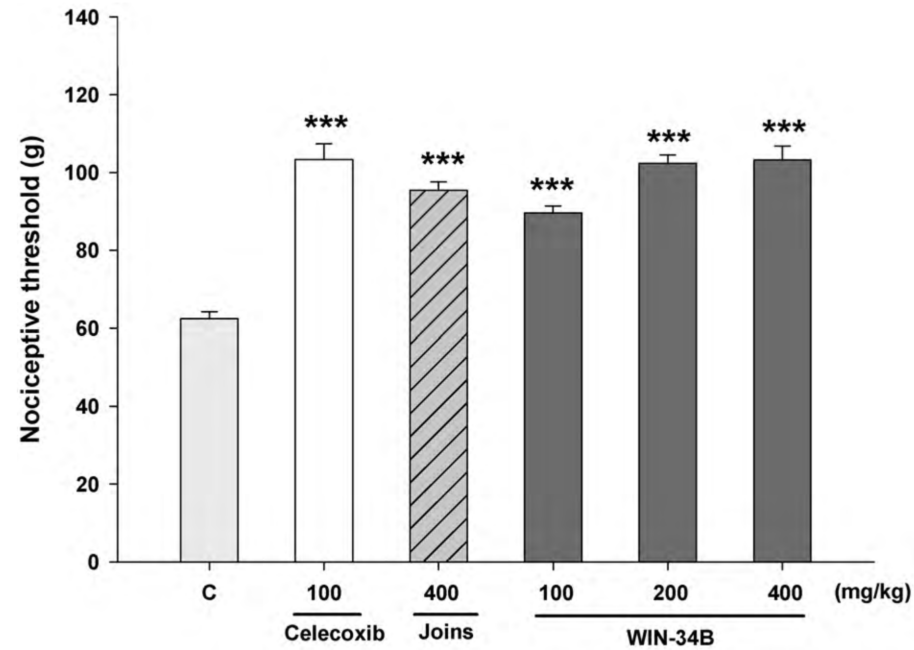

Fig. 4. Effects of WIN-34B on carrageenan-induced paw pressure in rats. The nociceptive threshold (g) was the force applied to dorsal surface of the inflamed hind paw, which caused each animal to withdraw the paw. The values are expressed as the mean ± S.E.M. ($n = 5$–9). $^{*}p < 0.05$, $^{**}p < 0.01$ and $^{***}p < 0.001$ indicate statistically significant differences from the vehicle control group.

WIN-34B at 400 mg/kg was very similar to that of celecoxib. This result suggests that WIN-34B has similar potency to celecoxib in pain relief.

3.9. Hargreves test

WIN-34B produced a significant reduction in thermal hyperalgesic effect induced by CFA 3 and 24 h after CFA treatment, as indicated by increased paw withdrawal latency (see Table 5, $p < 0.001$). Such effect of WIN-34B suggests it has higher potency than celecoxib and Joins^{TM}.

3.10. Acetic acid-induced vascular permeability

As shown in Fig. 5, WIN-34B at oral doses of 100, 200 and 400 mg/kg exhibited inhibition of 16.6%, 15.2% and 26% respectively in vascular permeability assay. Especially, WIN-34B at the oral dose of 400 mg/kg showed higher potency than that of Joins^{TM} (Joins at 400 mg/kg, 15.7%).

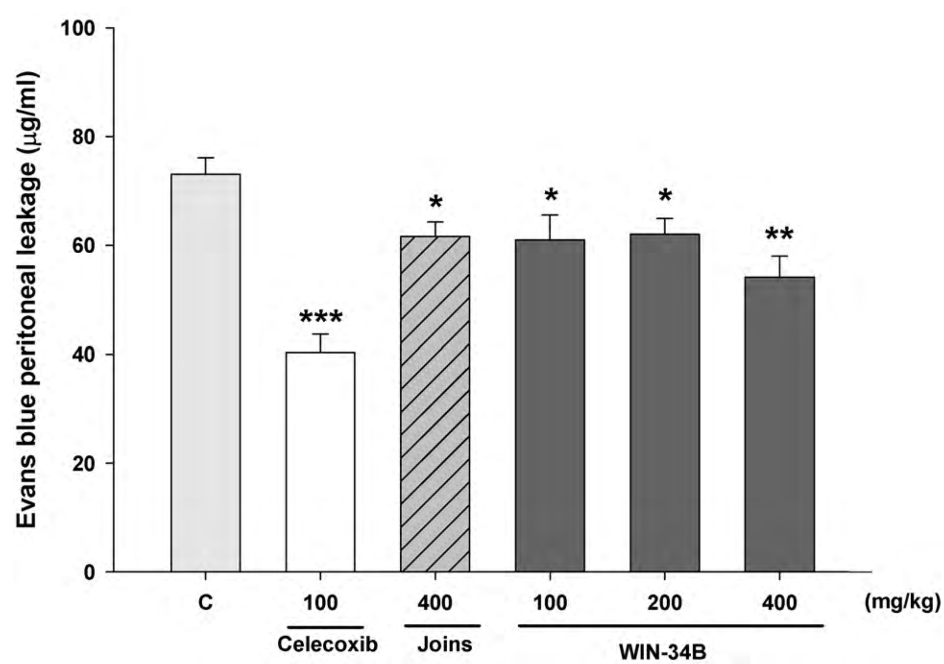

Fig. 5. Effects of WIN-34B on acetic acid-induced vascular permeability in mice. Dye leakage was determined 20 min after acetic acid injection. The values are expressed as the mean ± S.E.M. ($n = 6$–7). $^{*}p < 0.05$, $^{**}p < 0.01$ and $^{***}p < 0.001$ indicate statistically significant differences from the vehicle control group.

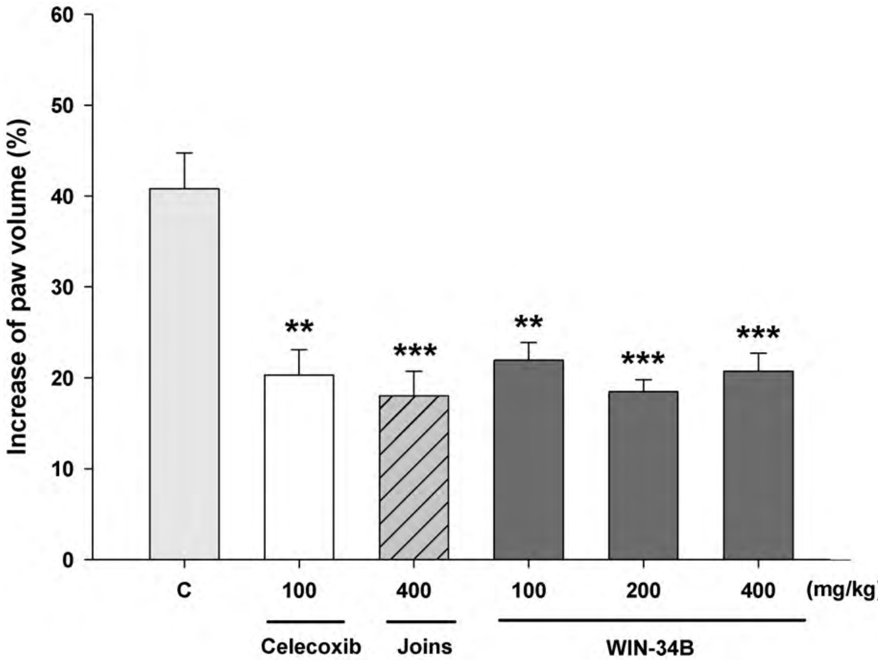

Fig. 6. Effects of WIN-34B on carrageenan-induced paw edema in rats. Increase of paw volume (%) was attained by following equation; $(a-b)/b100$. The values are expressed as the mean ± S.E.M. ($n = 7$). *$p < 0.05$, **$p < 0.01$ and ***$p < 0.001$ indicate statistically significant differences from the vehicle control group.

3.11. Carrageenan-induced paw edema

WIN-34B (100, 200 and 400 mg/kg) significantly inhibited development of carrageenan-induced paw edema after 3 h of treatment (see Fig. 6, $p < 0.001$). This inhibiting effect of carrageenan-induced inflammation by WIN-34B (400 mg/kg) was similar to that produced by a positive control celecoxib (100 mg/kg) ($p < 0.001$). However, WIN-34B at 200 mg/kg showed higher potency than celecoxib (54.8% *vs.* 50.3%, $p < 0.001$).

3.12. Croton oil-induced ear edema

Fig. 7 shows the anti-inflammatory effect of WIN-34B (200 and 400 mg/kg) on croton oil-induced ear edema in mice in dose-dependent manner. Celecoxib treatment significantly reduced ear edema formation (Fig. 7). However, WIN-34B at 400 mg/kg showed better anti-inflammatory effects than celecoxib at 100 mg/kg and Joins^{TM} at 400 mg/kg (39.2% *vs.* 31.5% *vs.* 24.4%, $p < 0.001$). This result suggests that WIN-34B at 200 and 400 mg/kg has better anti-inflammatory effect than that of celecoxib, a commonly used COX-2 inhibitor.

3.13. MIA-induced osteoarthritis

Experiments designed to determine the effects of WIN-34B on MIA-induced weight-bearing differential were conducted after MIA-induced osteoarthritis. WIN-34B significantly reduced change in hind paw weight distribution in a dose-dependent manner, 4, 7, 11 and 14 days post treatment (see Table 6, $p < 0.001$). Throughout these times, celecoxib produced a similar or inferior effect to WIN-34B at 400 mg/kg (see Table 6). This result suggests that WIN-34B at 400 mg/kg could be a superior anti-osteoarthritis agent, compared to celecoxib.

4. Discussion

The present study assessed the anti-nociceptive and anti-inflammatory activities of WIN-34B as a potential, novel treatment of osteoarthritis. Reduction in acetic acid-induced writhing responses, which is a pain model, may be related to reduce synthesis of inflammatory mediators by inhibition of cyclooxygenases and/or lipoxygenases (Franzotti et al., 2000; Ojewole, 2006). Results indicated that WIN-34B at 100, 200, and 400 mg/kg could decrease the number of writhes on animal model, therefore exhibited significant anti-nociceptive effects ($P < 0.01$) (Fig. 2). This result shows

Table 6
Effects of WIN-34B on MIA-induced osteoarthritis in rats.

Group	Dose mg/kg, p.o.	Change in hind paw weight distribution (%)				
		0 day	4 day	7 day	11 day	14 day
Control (0.5% CMC)	0	31.6 ± 1.1	27.5 ± 0.7	24.7 ± 0.7	22.8 ± 0.6	24.5 ± 1.0
Celecoxib	100	31.6 ± 1.0	29.3 ± 0.3*	29.8 ± 0.5***	29.9 ± 0.7***	28.5 ± 1.2*
Joins	400	31.7 ± 0.9	29.1 ± 0.7	28.1 ± 0.77**	25.5 ± 1.0*	25.9 ± 1.1
	100	31.7 ± 0.9	29.0 ± 0.7	26.9 ± 0.6*	24.0 ± 0.6	24.5 ± 1.0
WIN-34B	200	31.8 ± 0.8	30.9 ± 0.5***	27.4 ± 0.7*	26.3 ± 0.8**	26.8 ± 0.5*
	400	32.1 ± 0.9	31.4 ± 0.5***	29.4 ± 0.8***	29.8 ± 1.0***	30.5 ± 0.9***

The values are expressed as the mean ± S.E.M. ($n = 10$).
* $p < 0.05$ indicates statistically significant differences from the vehicle control group.
** $p < 0.01$ indicates statistically significant differences from the vehicle control group.
*** $p < 0.001$ indicates statistically significant differences from the vehicle control group.

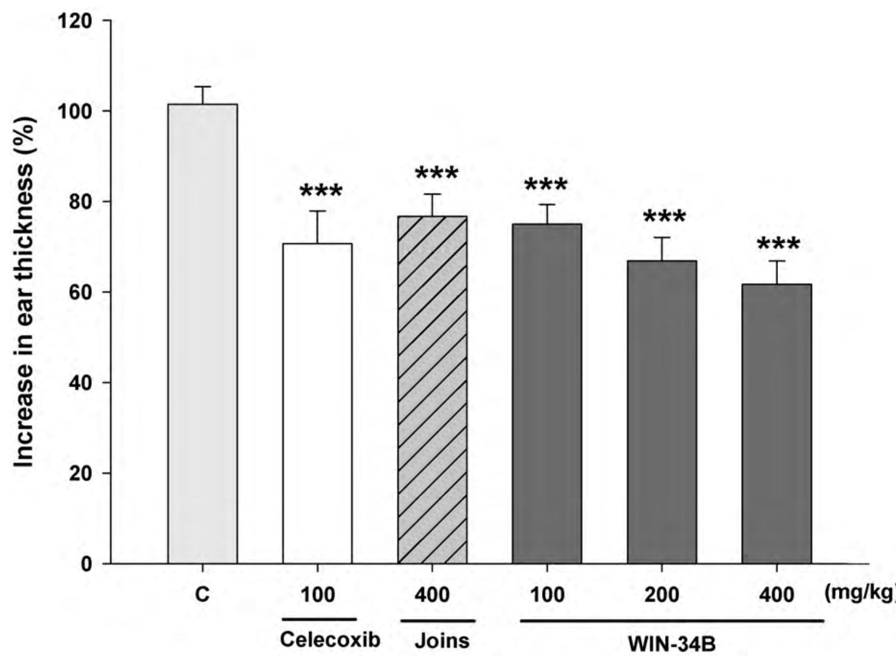

Fig. 7. Effects of WIN-34B on croton-induced ear edema in mice. Increase in ear thickness (%) was attained as it follows; {(Right ear thickness – Left ear thickness)/Left ear thickness} × 100. The values are expressed as the mean ± S.E.M. ($n = 12–13$). $^{*}p < 0.05$, $^{**}p < 0.01$ and $^{***}p < 0.001$ indicate statistically significant differences from the vehicle control group.

that WIN-34B produces significant analgesic effects, and is thought to be due to inhibition of arachidonic acid metabolite synthesis as well as the presence of biologically active chemical compounds in WIN-34B. In formalin test, WIN-34B at 200 and 400 mg/kg could reduce the duration of paw licking in the late phase (inflammatory) of the formalin test (Table 3). The significant effects of WIN-34B at 400 mg/kg ($p < 0.05$) and 200, 400 mg/kg ($p < 0.01$, $p < 0.001$) (Table 4) on hot plate responses at 60 and 120 min after treatments of drugs were shown. This result suggests that WIN-34B has a fast significant pain relief, while indomethacine and Joins™ did not show any significant pain relief at 60 min and maintains significant pain relief as much as indomethacine and Joins™ at 120 min. In radiant heat tail-flick test (see Fig. 3), WIN-34B significantly induced a dose-dependent increase of the latency time ($p < 0.001$). In carrageenan-induced paw pressure (see Fig. 4), WIN-34B significantly induced a dose-dependent increase of paw withdrawal threshold ($p < 0.001$). Such effect of WIN-34B at 400 mg/kg was very similar to celecoxib. It is known that sensitization of the nociceptors (hypernociception) induced by carrageenan has been a metabotropic event involving at least the stimulation of cAMP/PKA/Ca^{2+} or the phospholipase/PKC/Ca^{2+} pathways (Taiwo and Levine, 1989; Cunha et al., 1999; Ferreira and Nakamura, 1979; Lynn and O'Shea, 1998; Taniguchi et al., 1997; Coderre, 1992). Therefore, it may be considered that WIN-34B may produce an anti-nociception effect by interrupting such pathways. In CFA-induced paw thermal hyperalgesia (see Table 5), WIN-34B produced a significant reduction in thermal hyperalgesic effect at 3 and 24 h post CFA treatment (see Table 5, $p < 0.001$). WIN-34B at 400 mg/kg has higher potency than celecoxib and Joins™ producing such an effect. It has been reported that CFA produces long-term inflammatory and nociceptive responses through the NF-κB pathway (Chan et al., 2000). Therefore, WIN-34B may be considered to have anti-nociception effect by interrupting such pathways. On the other hand, to elucidate the anti-inflammatory property of WIN-34B, three *in vivo* models were used: vascular permeability, carrageenan-induced paw edema and croton oil-induced ear edema. In the early stage of inflammatory responses, the anti-inflammatory activity was assessed by inhibition of vascular permeability, itself a prominent feature of the inflammatory pathological process. Vascular permeability induced by acetic acid was measured by the amount of Evan's blue dye from the abdominal cavity (Miles and Miles, 1952). The increases of vascular permeability are attributed to leakage of plasma from the blood vessels in inflammatory swellings. This results in the dilation of arterioles and venules and increased vascular permeability (Lee et al., 2008). As demonstrated in Fig. 5, WIN-34B at 100, 200 and 400 mg/kg showed an inhibition of 16.6%, 15.2% and 26% in vascular permeability assay, respectively. The WIN-34B had significant anti-inflammatory effects on vascular permeability, which may in turn have an anti-inflammatory swelling effect that decreases capillary permeability. Carrageenan-induced paw edema as an *in vivo* model of inflammation has been frequently used to assess the anti-edematous effect of natural products. The inhibitory effect of carrageenan-induced inflammation by WIN-34B (200 and 400 mg/kg) was similar to that produced by the positive control, celecoxib (100 mg/kg) (see Fig. 6, $p < 0.001$). It is known that the third phase of the edema induced by carrageenan, in which the edema reaches its highest volume, is characterized by the presence of prostaglandins and other compounds of slow reaction (Spector and Willoughby, 1963). It is suggested that the anti-inflammatory mechanism of WIN-34B may be related to inhibition of prostaglandin synthesis, as described above (Di Rosa et al., 1971). Fig. 7 shows the anti-inflammatory effect of WIN-34B (100, 200, 400 mg/kg) on croton oil-induced ear edema in mice in a dose-dependent manner. WIN-34B shows strong inhibitory activity on croton oil-induced ear edema in mice, which is more pronounced than positive control celecoxib. Topical application of croton oil induces significant inflammatory responses characterized by oedema, neutrophil infiltration, prostaglandins production and increases in vascular permeability (Rao et al., 1993). The majority of the activities of croton oil appear to be involved or to be dependent on AA release and metabolism (Zhou et al., 2006). Therefore, the results of the present study suggest that WIN-34B may have pharmacological properties similar to both lipoxygenase and cyclooxygenase inhibitors. The monosodium iodoacetate (MIA) model of OA has been well described in rats, especially in terms of pathological progression of the disease, pain behavior and peripheral nerve sensitization. Injection of MIA into joints inhibits glyceraldehye-3-phosphate dehydrogenase activity in chondrocytes, resulting in disruption of glycolysis and eventual cell death (Kalbhen, 1987; van der Kraan et al., 1989). The progressive loss of chondrocytes results in histological and morphological changes in the articular cartilage, closely resembling those seen in human OA (Williams and Brandt, 1984). WIN-34B significantly decreased change in hind paw weight distribution in a dose-dependent manner at 4, 7, 11 and 14 days after treatment (see Table 6, $p < 0.001$) in the MIA-induced osteoarthritis animal model. In addition, celecoxib has a similar or inferior effect to WIN-34B at 400 mg/kg at 4, 7, 11 and 14 days after treatment (see Table 6). Those results suggest that WIN-34B might be a superior anti-osteoarthritis agent, compared to celecoxib through acute and chronic inflammatory pain relief and inflammation inhibition. The details of WIN-34B's anti-osteoarthritis mechanism are as follows: it is known that chlorogenic acid, the primary ingredient in *Lonicera japonica* Thunb, was shown to have anti-inflammatory effects (Qian et al., 2008). It has been reported that chlorogenic acid inhibited inflammation by suppressing LPS-induced COX-2 expression via attenuating the activation of NF-κB and JNK/AP-1 signaling pathways in Raw 264.7 cells *in vitro* (Shan et al., 2009). Chlorogenic acid inhibited carrageenan-induced paw edema and formalin-induced pain in rats *in vivo* (dos Santos et al., 2006). Mangiferin, which has strong anti-inflammatory properties, is a major active constituent of *Anemarrhena asphodeloides* BUNGE (Jung et al., 2009). It has been known that mangiferin inhibited LPS-induced nitric oxide and prostaglandin E_2 through NF-κB inactivation in RAW 264.7 macrophages and rat microglial cells, which exhibits anti-inflammatory effect *in vitro* (Bhatia et al., 2008; Garrido et al., 2004a,b). Two components, loniceroside C and anemarsaponin B of *Lonicera japonica* Thunb (LJT) and *Anemarrhena asphodeloides* BUNGE (AAB) have shown their anti-inflammatory effects (Kim et al., 2009; Kwak et al., 2003). However, it has been reported that the amounts of those two components in LJT and AAB are very small (Kim et al., 2009; Kwak et al., 2003). Therefore, it is thought that WIN-34B may work as an anti-osteoarthritic agent by decreasing inflammation and pain through inhibition of leukotrien and prostagrandin productions by main components such as chlorogenic acid and mangiferin, and unknown components.

In conclusion, WIN-34B exhibits better anti-nociceptive and anti-inflammatory activities than celecoxib or Joins^{TM}, even in osteoarthritic animal models. Furthermore, it exhibits better immediate analgesic effects than that of Joins^{TM} and celecoxib in the hot plate test (see Table 4), and exerts cartilage protection through protection of the cartilage matrix components based on our preliminary data. Meanwhile according to our preliminary data, it did not appear to exhibit any adverse gastrointestinal and cardiac effects. WIN-34B appears to have strong potential to become a novel therapy for osteoarthritis with a superior efficacy and safety profile when compared to other commonly used drugs, including Joins^{TM}, indomethacin, diclofenac and celecoxib. Pre-clinical research of WIN-34B has been already finished. Afterwards, a phase II clinical trial of WIN-34B is being conducted now in Korea after the IND filing of WIN-34B was approved by KFDA on July 24th, 2009. Good results are expected from this phase II clinical trial of WIN-34B. The IND filing of WIN-34B to other countries containing FDA is being prepared. In near future, it is strongly expected for FDA approval for the IND filing of WIN-34B.

Acknowledgement

This study was supported by a grant from the Oriental Medicine R&D Project, Ministry of Health and Welfare, Republic of Korea (B030008).

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